

AQUABOT: Components and Connection Detailing

1. Components List

Subsystem / Function	Component Specification
Locomotion Brain	Arduino Uno R3
Telemetry Brain	ESP32 NodeMCU Wi-Fi
Drivetrain	4WD Nezha Bot Chassis
Motor Interface	L298N / NeZha Expansion
Remote Guidance	2.4 GHz PS2 Controller
Autonomous Sensor	5-Channel IR Line Array
pH Payload	Glass Electrode Probe
Moisture Payload	Capacitive Soil/Liquid Sensor
Inspection Rig	Fixed Direct-Contact Arm
Visual Telemetry	16×2 I2C Character LCD
Alarm Suite	Red, Yellow, Green LEDs & Active Buzzer
Power Management	Rechargeable Battery

2. Connection Detailing / Pinout Mapping

Arduino Uno (Motor Locomotion & Navigation)

Pin / Connection	Component / Function
Pin 3 & Pin 5	Left and Right Motor Drive / PWM Speed Control
Pin A1 to A5	5-Channel Infrared Tracking Sensor Array (Far-Right to Far-Left)
SPI Pins (10, 11, 12, 13)	PS2 Controller Receiver Interface (CS, MOSI, MISO, SCK)
Pin 7 & Pin 6	SDA and SCL connected from the chassis board
Power	Power and GND connected from the chassis robot to the Arduino

ESP32 (Telemetry Hub & Signaling)

Pin / Connection	Component / Function
GPIO 34	Analog Input — Capacitive Moisture / Flow Presence Sensor
GPIO 32	Analog Input — Glass pH Sensor Signal Conditioner
GPIO 21 (SDA) & GPIO 22 (SCL)	16×2 I2C Character LCD Display
GPIO 25	Green Status Warning LED
GPIO 26	Yellow Status Warning LED

Pin / Connection	Component / Function
GPIO 27	Red Status Warning LED
GPIO 14	Active Buzzer Audio Output
Power	5V and GND connected from the chassis robot to the ESP32